EFY TESTED Ashwini K. Sinha

Voice Based
HOME AUTOMATION SYSTEM

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This project presents a voice based home automation system that can be used by differently abled people, geriatrics, and the others to control home appliances and thus convert a non-smart home into a smart home. To achieve this in a simple way, Amazon Echo

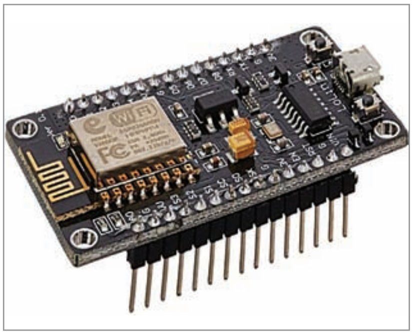

Fig. 1: NodeMCU controller board

Fig. 2: Alexa Echo Dot

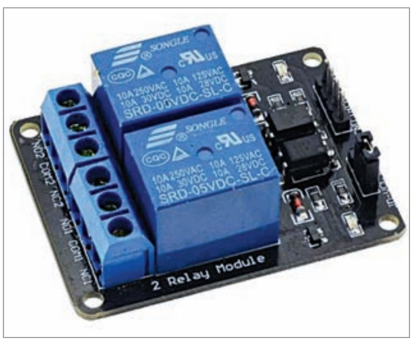

Fig. 3: Two-relay module

is used to communicate with the circuit to control the appliances.

The controller used for the automation in this project is NodeMCU. When a command is given by the user to Alexa, NodeMCU gets the information through Wi-Fi and decides the switching action to be taken with respect to the electrical appliances connected to it through relays.

Main components used in the project are shown in Fig. 1 through Fig. 3, while circuit diagram of the home automation system using Alexa is shown in Fig. 4. NodeMCU used as the controller board is preprogrammed to connect with the Wi-Fi network and then take the commands from Alexa to perform the tasks as per instructions in the code, like turning on or turning off the lights or fans.

Connect input pins IN1 and IN2 of the 2-channel relay to digital pins of the NodeMCU as per the code. If you are controlling pins D2 and D3 of NodeMCU using Alexa, then the relay IN pins should be connected to the same D2 and D3 pins.

BILL OF MATERIAL

Components	Description	Quantity
Amazon Alexa Eco Dot	Alexa based smart speaker	1
NodeMCU	Microcontroller board	1
5V, 2-relay module (R1, R2)	1A, 110V-220V AC relays	1
B1, B2	230V AC, 60W bulbs	2
Wires	Female to female jumper wires	4

The ground pin in NodeMCU and the relay are connected similarly; the Vcc pin of relay is connected to Vin of NodeMCU. Vcc and GND pins of NodeMCU are

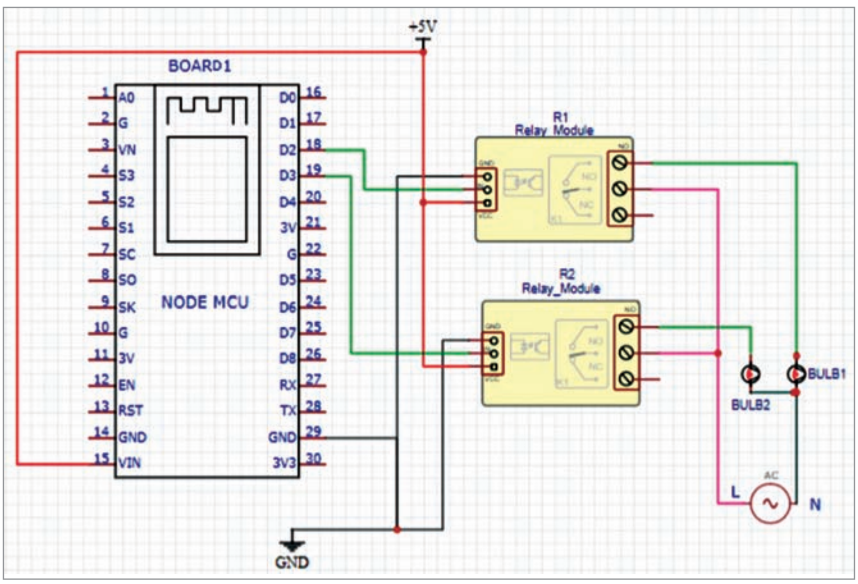

Fig. 4: Circuit diagram of the home automation system